

Kamran Amini

Senior Systems & Solutions Architect
Independent Consultant

- ✓ CTO Advisory & Strategic Decision Making
- ✓ Systems & Solutions Architecture
- ✓ Blockchain & AI Adoption
- ✓ Data & Network Security

Author of Books & Articles on Software Engineering

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SUMMARY

With over 15 years of experience across software development and systems architecture, I have worked across startups, large corporations, and consulting engagements as an engineer, Tech Lead, CTO, board member, and founder of my own software company, giving me first-hand insight into the technical, financial, and organizational dynamics of software firms at every scale. I combine hands-on engineering with architectural leadership, coding critical components myself so my architectural decisions stay informed by implementation reality. My core expertise lies in systems-level languages (C, C++, Golang, Rust), plus extensive DevOps experience with Kubernetes and AWS/GCP. In blockchain and fintech, I have architected tokenised-securities infrastructure (on-chain exchange and auction engines with atomic settlement, a central securities depository, brokerage and custody) alongside ERC-20/ERC-721 token and NFT platforms. More recently I build AI platforms end to end (self-hosted LLMs, vision and diffusion models, LoRA training) and work daily with AI coding agents (Claude, Codex, Gemini). Having witnessed organizational dysfunction firsthand, I bring that insight to the systems architecture, CTO-level strategy, and blockchain/AI consulting I provide today.

CAREER

DeKrypt B.V.

CEO & CTO

March 2024 - Present

The Netherlands, Amsterdam

- **Design & Architecture:** Building system foundations through deliberate decisions about data models, trust boundaries, and failure behavior rather than accident-prone defaults.
- **AI-Driven Development:** Leveraging agentic AI for scalable delivery while keeping engineers focused on the judgment and accountability that can't be delegated.
- **Distributed Ledger:** Designing systems where records remain verifiable by parties who may distrust the custodian.
- **FinTech Systems:** Architecting payment and settlement infrastructure compliant with industry standards, keeping books balanced even after failures.
- **Consultancy:** Providing architecture reviews and technical due diligence through adversarial analysis.
- **Writing & Research:** Publishing books, essays, and academic work that make the firm's expertise available for peer review and citation.

Vero Labs Inc.

Senior Blockchain Architect (Part-time consultancy)

April 2024 - September 2026

United States, New York

↳ Tokenise Ltd

Senior Blockchain Architect (Part-time consultancy)

United Kingdom, London

↳ VeroPulse GmbH

Chief Software Architect (Part-time consultancy)

Austria, Vienna

- **Agora Stock Exchange Engine:** Architected the on-chain orderbook and settlement engine, with atomic settlement finality enforced between the orderbook and treasury contracts.
- **Tokenise CSD:** Designed the Central Securities Depository platform responsible for on-chain issuance, custody, and settlement of tokenized securities.
- **Tokenise Broker:** Architected the brokerage platform connecting investors to the Agora Engine for order routing and execution.
- **Vero Android App:** Directed the architecture of the primary and secondary market mobile client, integrated end-to-end with the Agora Engine.
- **Platform & Infrastructure:** Designed the off-chain microservices layer in Golang, deployed on Kubernetes (AWS EKS) for scalable, highly-available trading workloads.
- **AI-Assisted Engineering:** Introduced AI-assisted coding workflows (LLM copilots, automated code review) across the teams to accelerate delivery and improve code quality.
- Golang - Kubernetes (EKS) - AWS - Solidity - EVM chains - PostgreSQL - Docker - gRPC

Assetera GmbH

Senior Blockchain Architect (Part-time consultancy)

April 2024 - April 2026

Austria, Vienna

- **Agora Auction Room:** Architected the on-chain auction mechanics, with atomic settlement finality enforced against the on-chain treasury.
- **Tokenisation as a Service:** Architected the on-chain asset tokenisation framework, covering issuance, custody, and transfer of tokenized assets.
- **Platform & Infrastructure:** Designed the off-chain Golang services, deployed on Kubernetes (AWS EKS).
- Golang - Solidity - EVM chains - Kubernetes (EKS) - AWS

Digital First GmbH

Senior Blockchain Architect (Part-time consultancy)

December 2023 - September 2026

Austria, Vienna

- **TheKiss NFT Tokenization Platform:** Designed the NFT minting architecture and smart contract framework underpinning the platform's digital collectibles.
- **ARTEO Token:** Architected the ERC-20 token economics and smart contract suite for the art-backed digital asset.
- **QLINDO Token:** Led smart contract architecture and security review for the project's token launch.
- **Hergé NFTs:** Architected the ERC-721 minting and royalty framework for the licensed NFT collection.
- Solidity - EVM chains - ERC-20 - ERC-721 - Hardhat - OpenZeppelin

ActiveVideo B.V. (April 2019 - February 2023)

↳ Commscope Inc. (March 2023 - March 2024)

↳ ARRIS B.V. (April 2024 - December 2025)

↳ Vistance Networks B.V. (January 2026 - Present)

Senior Software Engineer

The Netherlands, Hilversum

- **CloudTV (Video Stitching Engine):** Cloud HTTP renderer/streamer broadcasting menus, UI, and audio/video to set-top boxes, built on Java, C++, WebKit, and Perl.
- **AppCloud:** On-demand containerized Android streaming platform, with its microservices backend deployed on Kubernetes (AWS EKS).
- **Multiview Engine:** Cloud HTTP renderer/streamer broadcasting multiple channels simultaneously to set-top boxes and TVs.
- **Member of Tech Council:** Prioritized tech debt against product objectives via a dedicated backlog and Kanban flow.
- **Founding member of Security Guild:** Devised and enforced security standards across teams, and supported product certification.
- C++17 (g++, clang) - Golang - Java, and Kotlin - Python - AWS - Kubernetes (EKS) - S3 - EFS - EBS - Ceph - Envoy

Metapolitan Ltd.

Chief Blockchain Architect (Part-time consultancy)

January 2025 - March 2026

United Kingdom, London

- **Distributed Treasury System:** Architected a multi-signature, on-chain treasury platform for fintech clients, enabling secure custody, automated settlement, and policy-driven fund controls across multiple blockchain networks.
- Solidity - Gnosis Safe (multi-sig) - EVM chains - web3.js - Hardhat - Node.js

TomTom B.V.

Senior Software Engineer

September 2017 - March 2019

The Netherlands, Amsterdam

- Expert Software Consultant from Modis B.V.
- NavKit Free Text Search Engine, Navkit Worker REST Services, POI Info Services
- C++11 (g++, clang) - Java SE - JUnit - Spring MVC - JAXB - Spring REST - gdb - Valgrind - Clang ASAN & TSAN - STL & Boost - Google Test - Protobuf - Thrift - Python - Jenkins - QuickBuild - Docker - JIRA - Fisheye - Confluence - Bitbucket Stash - AWS/Azure based cloud

Jeppesen Systems (a Boeing subsidiary)

Senior Software Engineer

February 2017 - August 2017

Sweden, Gothenburg

- Manpower Application
- C++11 (g++) - Python - C#.NET - gdb - Valgrind (callgrind, massif) - PDB - STL & Boost - Google Test - XMLRPC - Jenkins - JIRA - Fisheye - Confluence

Adro Co.

Senior Software Architect, Principal C++ Engineer

June 2016 - January 2017

Iran, Tehran

- Realtime Bidding (RTB) Engine
- C++ (g++) - Python - gdb - Valgrind - STL & Boost - libuv - Google Test - Protobuf - Jenkins - Travis CI - Docker - JIRA - Fisheye

Naeem Co.

Chief Technical Officer, Chief Software Architect

April 2015 - October 2016

Iran, Tehran

- **Projects:** ePassport Issuance System - Face Recognition System, Biometric Identification Sys.
- **Cryptography:** OpenSSL - PKCS (1, 7, 10, 11, 12) - ICAO MRTD standards - HSM - Key Mgmt. - PKCS11 Tokens
- **Stack:** C (gcc) - C++ (g++) - Java SE - Java EE - Spring MVC - Wildfly - Tomcat - Python - gdb - Valgrind - STL & Boost - wxWidgets - Qt - Google Test - Jenkins - Docker - vSphere - Proxmox - Gluster - Cassandra - Redmine - GitLab

Sharif CERT Center

Software Engineer

December 2011 - June 2013

Iran, Tehran

- Botnet Detection System - Sinkhole - DDNS as a FreeBSD module
- C++ (g++) - Java SE - Perl - gdb - JConsole - Valgrind - STL & Boost - Swing - Redmine

TECHNICAL SKILLS

- **Languages & Architecture:** C++ (14/17/20), C, Golang, Rust, Python, Java, JS - Distributed systems (CAP, replication, leader election, CQRS, event sourcing, message queues, streaming, async/event loops)
- **Cloud & DevOps:** Kubernetes (kubectl, Helm, Kustomize, PSA) - AWS & GCP (EKS/GKE, EC2/GCE, S3, VPC, IAM) - Docker - PostgreSQL, MySQL, MongoDB, Redis
- **Blockchain:** Solidity (EVM chains), Solana, web3.js/ethers.js, Hardhat - Distributed ledgers, smart contract architecture
- **Security & Cryptography:** OpenSSL, PKI, symmetric/asymmetric algos, hashing/MAC - Protocols & standards (gRPC, Protobuf, REST, PKCS, RFCs) - Networking (TCP/IP, Wireshark, tcpdump) - CISSP (in progress)
- **AI & LLMs:** Self-hosted LLM/VLM & diffusion pipelines (Qwen, FLUX, Diffusers, Ollama) - LoRA training, identity conditioning (PuLID/IP-Adapter, face embeddings), inpainting, prompt compilation, agent memory, elastic GPU serving - Agentic coding & review with Claude, Codex, and Gemini
- **Other:** Linux kernel modules & system calls - Machine learning (TensorFlow, OpenCV, CUDA)

Technical authorship & open source projects

- **The Software Ecology** (to be published): Examines the software industry as a socio-technical ecology, where companies, teams, and regulators interact through human behavior, power, and organizational dynamics as much as technology.
- **Extreme C:** Book published October 2019 by Packt Publication.
- **Linux Two-Step Authentication with Rust:** liveProject for Manning Publications, with its accompanying ESS backend and client projects in Rust.
- **Other Rust projects:** Domain Catcher Engine and the Outlook .msg parser.

FOREIGN LANGUAGES

Persian (**Native**), English (**Fluent**), German (**A1**), Dutch (**B1**), French (**A1**)

EDUCATION

- (2017 - 2022) **Bachelor of Science in Physics (Astronomy and Planetary Science)** at The Open University
- (2004 - 2008) **Bachelor of Software Engineering** at University of Isfahan (Isfahan, Iran)

PERSONAL INTERESTS

- **Physics and Astronomy:** Cosmology, Astronomy, Quantum Physics, Quantum Mechanics, Particle Physics, Special and General Relativity, Theory of Everything, Photonics.
- **Mathematics:** Linear and Abstract Algebra, Probability Theory, Information Theory
- **Cryptography:** Information Security, Network Security, Security Algorithms, Quantum Information Theory, Quantum Key Distribution, Quantum Cryptography
- **Philosophy:** Western Philosophy, Sociology, Economics, Politics and Humanities
- **Telecommunications:** Modulation, Multiplexing, Photonics, optical fibers, IEEE 802.3, Coding Theory